Prior art Search Report

X-MPFD-E++: DETERMINISTIC MULTIMODAL FUSION ARCHITECTURE FOR ZERO-DAY PHISHING DETECTION AND FEATURE PROVENANCE TRACKING

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SECTION1A - Introduction

Intellectual property (IP) competitive analysis is a vital tool for companies seeking to develop and protect their technology innovations in a highly competitive market. Web security and phishing detection systems combining URL and lexical analysis, HTML/DOM structure inspection, visual and TLS certificate analysis, policy-driven fusion of detection results and auditable decision records have become increasingly important in various industries, including enterprise security, e-mail security, financial services and managed security operations. As a result, companies in this space are investing heavily in research and development to gain a competitive advantage. This IP competitive analysis enables a company to identify key patents and other intellectual property assets held by their competitors, as well as potential infringement risks and licensing opportunities. This analysis is crucial for a company to make informed decisions about their R&D investments, patent filings, and overall IP strategy, and to stay ahead in the fast-paced field of phishing and web threat detection. The search was performed using combinations of keywords mentioned below, patent classes, and assignees relevant to the domain. Further, the records have been categorised into different technical categories.

SECTION 1B–KEY FEATURES OF THE INVENTION

a deterministic multimodal phishing detection system, comprising:

1) Multimodal Evidence Acquisition (URL/Lexical, HTML/DOM, Visual and TLS Signals)
2) Deterministic Policy-Driven Fusion of Extracted Features
3) Conditional / Staged Invocation of Costly Analysis (Early Exit)
4) Structured Explanation Linked to Decision Evidence
5) Feature-Level Provenance with Integrity-Protected, Replayable Decision Record

Relevant References:

Patents / Literature	**1**	**2**	**3**	**4**	**5**
1. US11470114B2	√	√	√		
2. US9578048B1	√	√			
3. US11381597B2	√		√		
4. US12413621B2	√		√		
5. US20200036751A1	√		√		
6. US11438377B1	√		√		
7. Multimodal Phishing Website Detection System Using XAI (MAKE 2026)	√		√	√	
8. WebGuard++ (arXiv:2506.19356)	√			√	

Keywords (with variations):

· Phishing Detection / Malicious URL / Zero-Day Phishing

- Multimodal / Multi-Engine Fusion (URL, HTML/DOM, Screenshot, TLS Certificate)
- Rule-Based / Policy-Driven / Decision Tree Classification
- Staged Analysis / Early Exit / Rough Analysis / Conditional Rendering
- Missing Modality / Partial Modality / Fault Tolerance
- Explainable / Interpretable Phishing Detection / Evidence Package
- Feature Provenance / Decision Lineage / Tamper-Evident Record / Replay

In addition to manual and keyword searching in the subclasses above, keyword searching was conducted with no class limitation.

SECTION 2A – Search Results

results of the search -Relevant

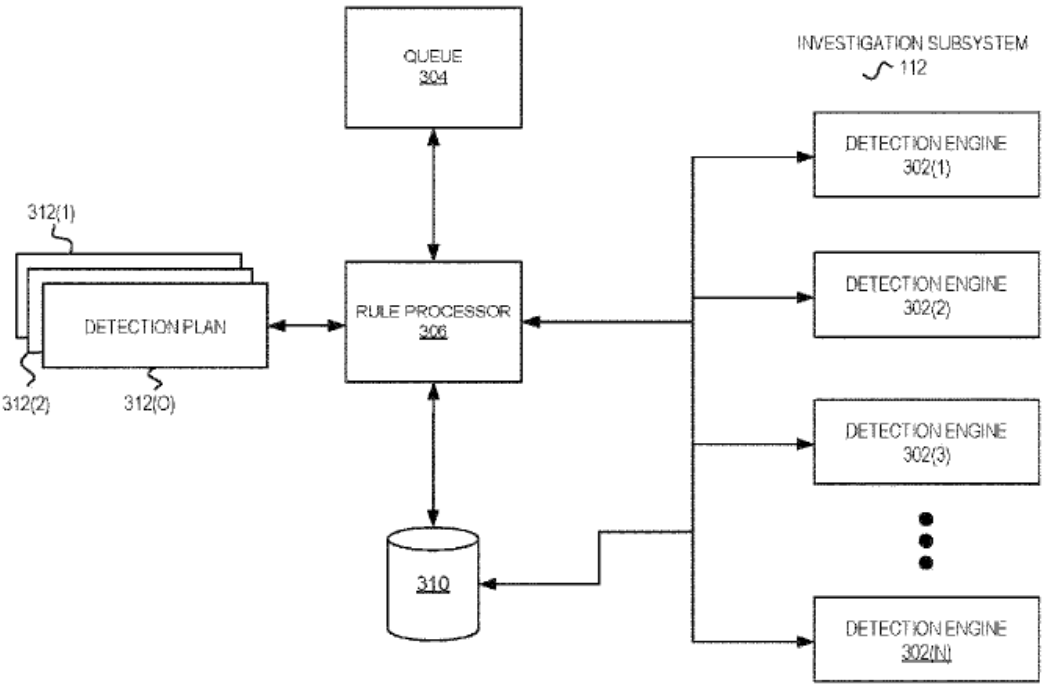
1. US11470114B2 MALWARE AND PHISHING DETECTION AND MEDIATION PLATFORM

[ABSTRACT]

A method for using a malware and phishing detection and mediation platform is discussed. The method includes accessing data from one or more of a monitored portion of website data and a monitored portion of emails, the data indicating a respective potential malware or a suspect phishing element (e.g., Uniform Resource Locator (URL)). The method includes selecting one of a plurality of detection engines for processing the data, where the selecting is based on previous results of previous processing by one or more detection engines. Each of the plurality of detection engines can be for performing one or more respective investigation actions on the plurality of data to determine a particular issue with one of the monitored data. The method also includes determining a mediation action based on a result of processing of the detection engine and the previous processing.

[DETAILD DESCRIPTION]

The document describes a malware and phishing detection and mediation (MAPDAM) platform 102 that receives potential phishing indicators, such as suspect URLs, from detectors 104 monitoring website data and e-mails, and processes them through ingestion, detection and action stages. At the detection stage, a rule processor 306 of an investigation subsystem 112 follows a configurable detection plan 312 that decides which of a plurality of detection engines 302 is run next, such as blacklist, URL rule and uniqueness scoring, website certificate and web page content modelling engines, the choice depending on results already produced by earlier engines. The rule processor 306 can skip remaining engines, for example when a URL is whitelisted, and can run certificate analysis only for URLs whose prior score indicates high phishing potential, each engine being able to continue analysis, suspend processing or initiate mitigation. Engine results are combined into a final determination, after which an action subsystem selects a mediation action and packages evidence captured during detection into communication packets for mitigation services such as safe browsing lists and hosting providers.



2. US9578048B1 IDENTIFYING PHISHING WEBSITES USING DOM CHARACTERISTICS

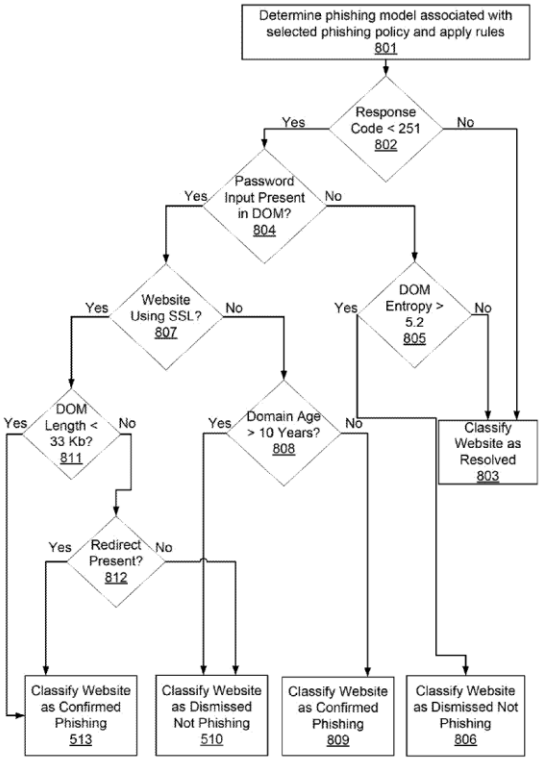
[ABSTRACT]

Embodiments of the present invention are directed to identifying phishing websites by rendering and analyzing document object model (DOM) objects associated with a website for features that indicate phishing behavior. Embodiments analyze the full scope and functionality associated with a website by executing functions embedded in a DOM object before analyzing the website for phishing activity. Accordingly, embodiments render and analyze a fully executed DOM object for phishing behavior. Embodiments may then perform steps to mediate a website that is classified as performing phishing. Thus, embodiments are configured to (1) collect website information from a variety of websites and web servers connected to the internet, (2) analyze the collected data to determine whether the website information is performing phishing, and (3) mediate websites and other actors that are determined to be performing phishing based on the results of the phishing analysis.

[DETAILD DESCRIPTION]

The document describes a computing device 210 having a DOM rendering module 213, a feature extraction module 214, a phishing model analysis module 215 and a phishing website mediation module 216, coupled to a phishing model database 217 and a phishing analysis results database 218. Website information received from a web server is built into a document object model (DOM) object and embedded JavaScript functions are executed, so that a fully executed DOM rather than static HTML is analysed. Non-DOM features, such as response status code, response body size, use of SSL, domain age and domain and path entropy and length, and DOM features, such as password and username form fields, third-party links and requests, redirect counts, DOM length, DOM entropy and keywords, are extracted. A phishing policy applies one or more phishing models, each a hierarchical decision logic of nodes holding conditional phishing rules

(FIG. 8) traversed to a final node that classifies the website as confirmed phishing, dismissed or resolved, each path carrying a path accuracy. Results of several models may be combined by majority vote, score threshold or weighting, and confirmed phishing websites are reported for take-down.



3. US11381597B2 EXPEDITION OF WEB PHISHING DETECTION FOR SUSPICIOUS SITES

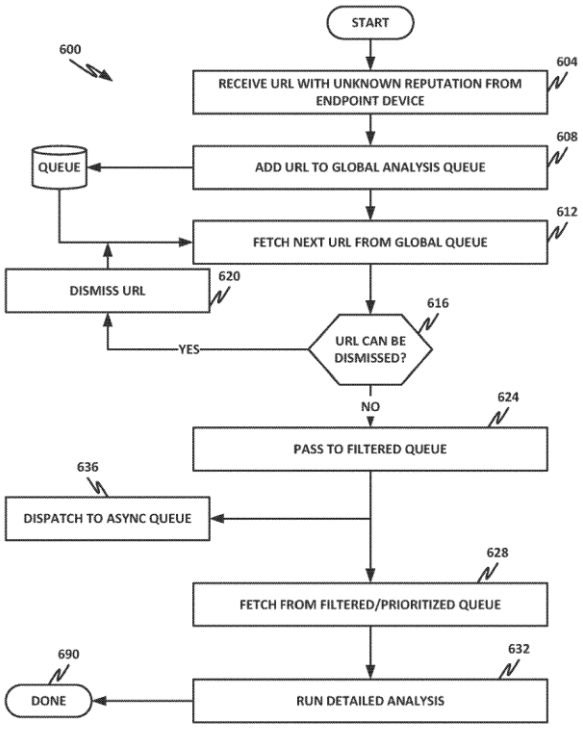
[ABSTRACT]

There is disclosed in one example a computing apparatus, including: a processor and a memory; a uniform resource locator (URL) reputation store; a network interface; and instructions encoded within the memory to instruct the processor to: receive via the network interface a request for a reputation for a URL; query the URL reputation store and determine that the URL does not have a known reliable reputation; add the URL to a URL analysis queue; perform a rough analysis of the URL, and determine from the rough analysis that the URL potentially is for a phishing website; and move the URL ahead in the analysis queue.

[DETAILD DESCRIPTION]

The document describes a URL reputation server 400 in which a filter module 404 first queries a global reputation store 408 and returns any known reputation, and otherwise applies filters such as absence of HTTPS, long URL paths, URL age, prevalence and campaign context. A rough analysis module 412 then performs a fast first-pass analysis by taking a low-resolution screenshot of the unknown website, redacting text areas by optical character recognition and comparing gross visual features against stored images of the top n phishing target websites held

in a phishing target store 416 (FIG. 7), the rough analysis being tuned for a low false negative rate at the cost of false positives. Where a match is found, a queue reordering module 420 moves the URL ahead in an analysis queue 424, optionally weighted by degree of suspicion and subject to a maximum timeout for other URLs, so that an analysis module 428 runs the slower detailed workflows on likely phishing URLs first (FIG. 6), reducing user exposure to zero-day phishing websites.



4. US12413621B2 VISUAL DETECTION OF PHISHING WEBSITES VIA HEADLESS BROWSER

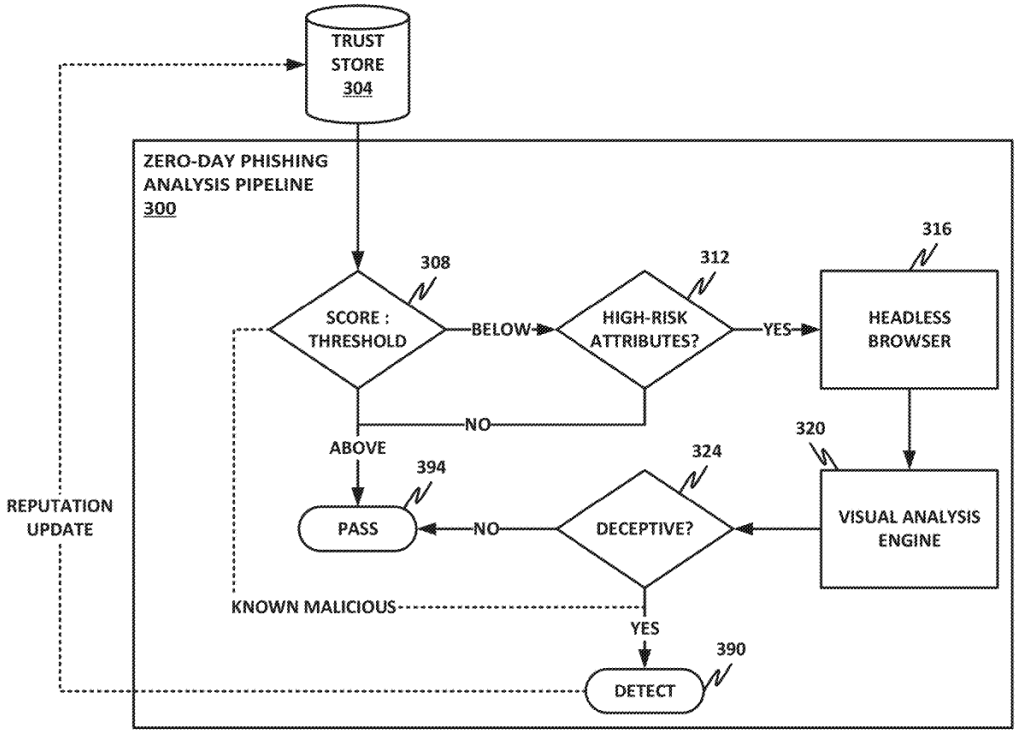
[ABSTRACT]

There is disclosed a computer-implemented system and method for protecting a user from phishing attacks, including detecting that the user has opened a website associated with a uniform resource locator (URL), and determining that the URL does not have a known good reputation; rendering a test version of the website in a headless web browser, including abstracting the website into at least one visual element of the test version; visually inspecting the test version of the website with a digital eye, and determining that the test version of the web looks like a known legitimate website not associated with the URL; and based on the visual inspection, warning the user that the website is or may be a phishing website.

[DETAILD DESCRIPTION]

The document describes a zero day phishing analysis pipeline 300 (FIG. 3) in which a trust store 304 is first queried for a reputation score of a URL opened by a user. A URL known to be malicious is detected at once, and a URL scoring above a trust threshold is passed without

further analysis, which the document identifies as the most common path. Only a URL lacking a reliable reputation proceeds to a filter 312 that checks for high-risk attributes, such as use of HTTP instead of HTTPS and the presence of input fields, a website without input fields being passed. A URL meeting both conditions is rendered in a headless browser 316, the rendered page is abstracted into visual elements such as logos, images and fonts, and a visual analysis engine 320 having a trained digital eye determines whether the page looks like a known phishing target. A deceptive page is detected and a reputation update is returned to the trust store 304, and the pipeline may be deployed in an e-mail server or an enterprise gateway (FIGS. 4 and 5).



5. US20200036751A1 PHISHING DETECTION SYSTEM AND METHOD OF USE

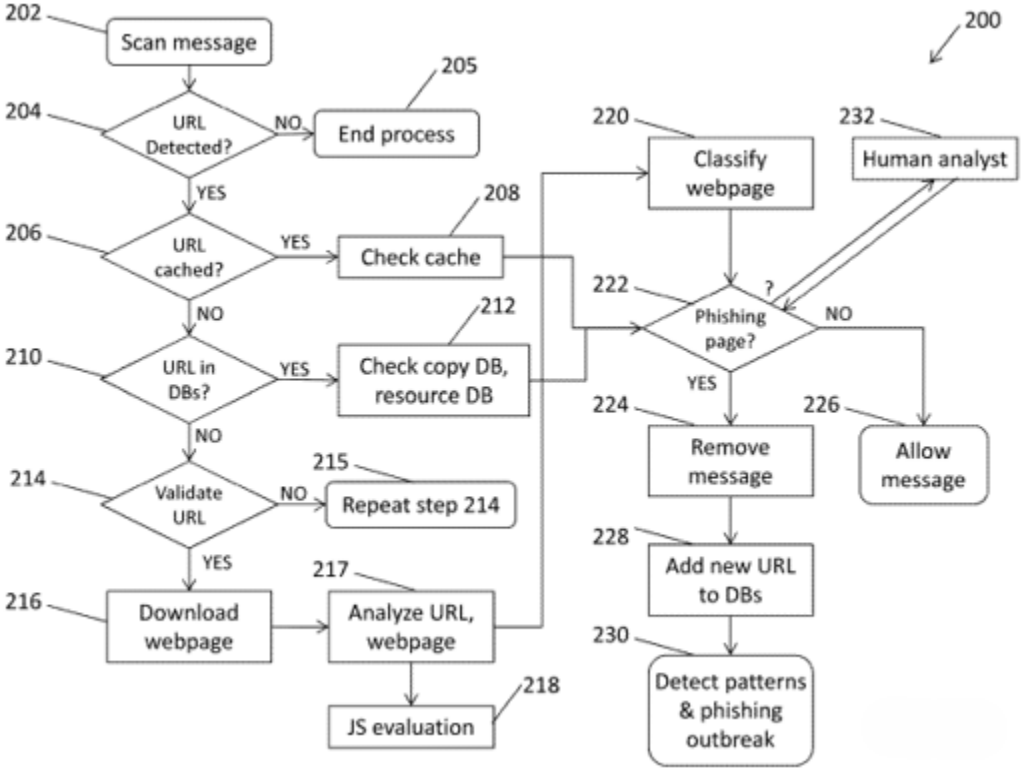
[ABSTRACT]

A system and method for detecting a phishing message comprises: providing a phishing detector comprising a scan engine and a fetcher; detecting a URL in the message by the scan engine; resolving the URL to a webpage by the scan engine; downloading the webpage by the fetcher; analysis of the downloaded webpage by the fetcher to determine whether the webpage is a phishing webpage; wherein when the webpage is determined to be a phishing webpage, deleting the message by the scan engine.

[DETAILD DESCRIPTION]

The document describes a phishing detection system 100 comprising a service provider 60 holding a resource database 132 and a central phishing analyser 138, and a local phishing detector 120 within a private messaging network 110 having a scan engine 122, a fetcher 124 with fetcher processor 126, a copy database 128 and a cache 129. In process 200 (FIG. 2), the scan engine 122 detects a URL in a message and first checks the cache 129 and then the

databases 128, 132 for an existing classification, which the document states uses fewer resources than full site analysis. Only an unclassified URL is validated, followed through link pages 152 and downloaded using the recipient IP range and several user agents. The URL and webpage are then analysed by phishing score contributors, including URL characteristics and WhoIs metadata, domain reputation, HTML element fingerprints, favicon, screenshot, CSS and logo comparison, phishing kit code, credential forms and language analysis. The scores are totalled to classify the webpage, a doubtful result is referred to a human analyst 232 who updates the rules, and a phishing message is removed from the user inbox.



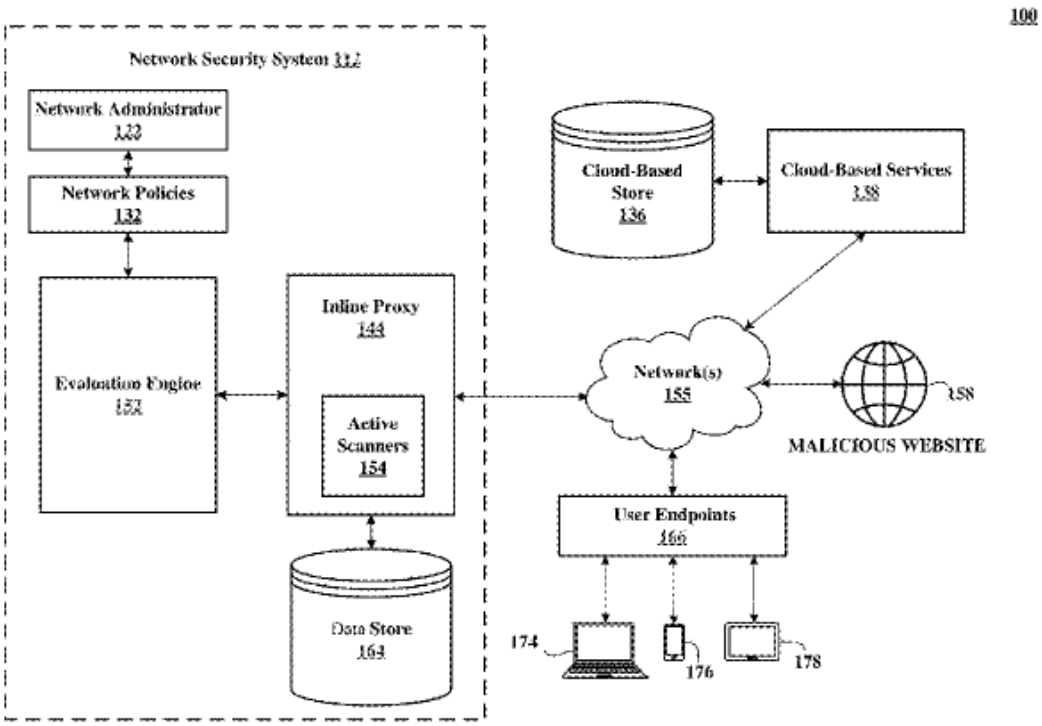
6. US11438377B1 MACHINE LEARNING-BASED SYSTEMS AND METHODS OF USING URLS AND HTML ENCODINGS FOR DETECTING PHISHING WEBSITES

[ABSTRACT]

Disclosed is classifying a URL and a content page accessed via the URL as phishing or not. URL embedder extracts characters in a predetermined set from the URL to produce a character string trained using ground truth classification of the URL, producing a URL embedding. HTML parser accesses content at the URL and extracts HTML tokens from the content page. Further, HTML encoder, trained on HTML tokens extracted from content pages at example URLs, each example URL accompanied by a ground truth image captured from the content page accessed via the example URL, produces an HTML encoding of the tokens extracted from the page. Also, phishing classifier layers, trained on the URL embedding and the HTML encoding of example URLs, processes a concatenated input of the URL embedding and the HTML encoding to produce a likelihood score that the URL and content accessed via the URL presents a phishing risk.

[DETAILD DESCRIPTION]

The document describes a network security system 112 having an inline proxy 144, active scanners 154 and an evaluation engine 152, which classifies a URL and its content page as phishing or not using deep learning. Three generations of phishing detection engine are disclosed. A first generation concatenates an encoding of web page text from a BERT encoder with an embedding of a captured page image from a ResNet model. A second generation trains an HTML encoder in an encoder-decoder pair whose decoder reconstructs the rendered page image from the HTML embedding, after which the decoder is replaced by classifier layers, optionally concatenated with image and URL feature embeddings. A third generation combines a URL embedder and URL feature hasher with the HTML encoder and phishing classifier layers to produce a likelihood score in real time. The document states that the faster third model can filter suspicious URLs without visiting the website, and that suspicious URLs can later be routed to the first or second engine for final detection, phishing URLs being blocked through the inline proxy 144.



7. A MULTIMODAL PHISHING WEBSITE DETECTION SYSTEM USING EXPLAINABLE ARTIFICIAL INTELLIGENCE TECHNOLOGIES (Alexey Vulfin et al., Machine Learning and Knowledge Extraction, Vol. 8, Issue 1, Article 11, 2026, DOI: 10.3390/make8010011)

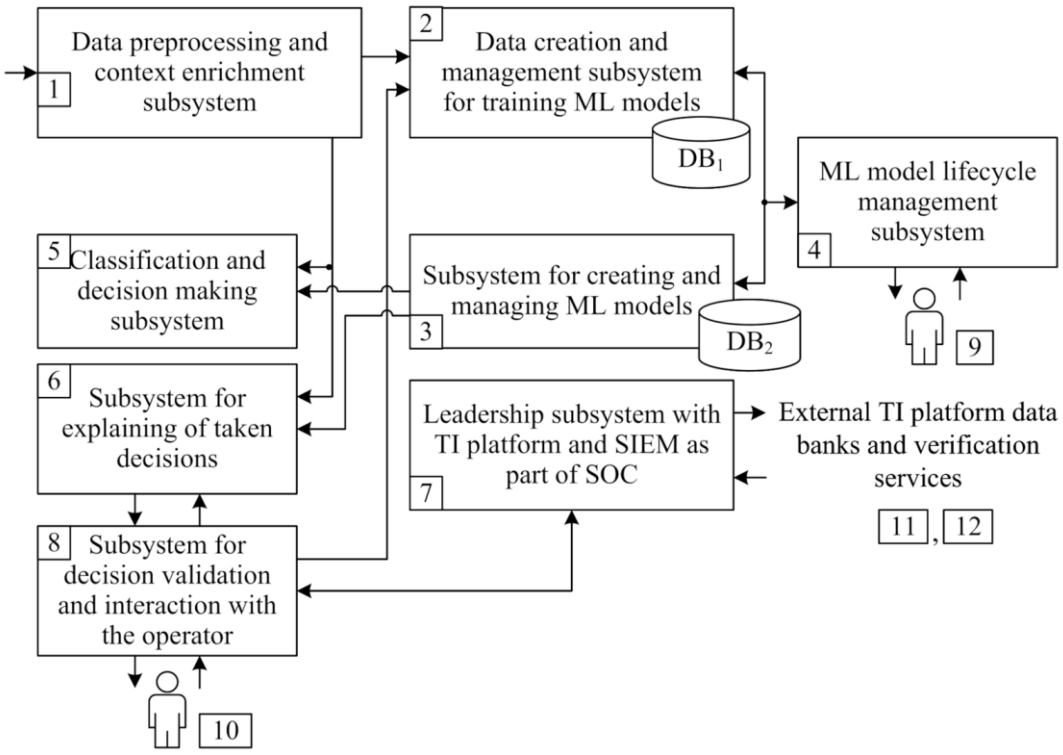
[ABSTRACT]

The purpose of the present study is to improve the efficiency of phishing web resource detection through multimodal analysis and using methods of explainable artificial intelligence. We propose a late fusion architecture in which independent specialized models process four modalities and are combined using weighted voting. The first branch uses CatBoost for URL features and metadata; the second uses CNN1D for symbolic-level URL representation; the third uses a

Transformer based on a pretrained CodeBERT for the homepage HTML code; and the fourth uses EfficientNet-B7 for page screenshot analysis. SHAP, Grad-CAM, and attention matrices are used to interpret decisions; a local LLM generates a consolidated textual explanation. A prototype system based on a microservice architecture, integrated with the SOC, has been developed. This integration enables streaming processing and reproducible validation. Computational experiments using our own updated dataset and the public MTLP dataset show high performance: F1-scores of up to 0.989 on our own dataset and 0.953 on MTLP; multimodal fusion consistently outperforms single-modal baseline models. The practical significance of this approach for zero-day detection and false positive reduction, through feature alignment across modalities and explainability, is demonstrated. All limitations and operational aspects (data drift, adversarial robustness, LLM latency) of the proposed prototype are presented. We also outline areas for further research.

[DETAILD DESCRIPTION]

The paper describes a web resource analysis system (Figure 5) having subsystems for data pre-processing and context enrichment, dataset and model management, classification and decision making, explanation, threat intelligence and SIEM integration, and analyst decision verification. The target URL, home page HTML and screenshot are processed by four independently trained models: CatBoost on more than 89 URL and metadata features, a character-level CNN on the URL, a CodeBERT transformer on HTML code and EfficientNet-B7 on the screenshot. A trainable meta-classifier combines their probabilities by late fusion and continues to operate on a reduced set of modalities when a screenshot or HTML cannot be obtained, the authors noting that fast URL branches can give a baseline result while heavier models are reserved for the most suspicious events. SHAP, Grad-CAM, integrated gradient and attention outputs are converted into a common representation and aggregated by a local LLM into a JSON class label and a templated text explanation for SOC analysts, and results are stored in MongoDB and Elasticsearch with logs that may be signed for investigation.



8. WEBGUARD++: INTERPRETABLE MALICIOUS URL DETECTION VIA BIDIRECTIONAL FUSION OF HTML SUBGRAPHS AND MULTI-SCALE CONVOLUTIONAL BERT (Ye Tian et al., arXiv:2506.19356, 2025)

[ABSTRACT]

The paper addresses four shortcomings of prior URL and HTML fusion approaches to malicious URL detection: incomplete URL modelling, dilution of sparse threat signals in HTML graphs, one-directional cross-modal analysis and opaque decisions without attribution to DOM components. It proposes WebGuard++, comprising a cross-scale URL encoder built on a transformer with dynamic convolution, a subgraph-aware HTML encoder that splits DOM graphs into interpretable substructures, a bidirectional coupling module that aligns URL and HTML embeddings by cross-modal contrastive learning, and a voting module that localises malicious regions from subgraph predictions. The authors report a true positive rate 1.1 to 7.9 times higher than baselines at false positive rates of 0.001 and 0.0001 on two datasets.

[DETAILD DESCRIPTION]

The paper describes a framework (Fig. 1) in which a cross-scale URL encoder combines CharBERT and ConvBERT, takes embeddings from all twelve ConvBERT hidden layers and fuses them through a spatial pyramid attention module. The HTML document is parsed with Beautiful Soup into a DOM tree, converted into a graph whose nodes carry Word2Vec embeddings of tag, attribute and text content, and partitioned by a hash function into subgraphs that are processed in batches by a graph neural network with node pooling, so that sparse malicious nodes are not diluted by benign content. A bidirectional coupling module stacks self-attention and cross-attention layers between URL and HTML features. A biased voting mechanism (Algorithm 1, Fig. 2) samples subgraph batches over several rounds and classifies

the URL-HTML pair as phishing when more than one round predicts malicious, the flagged subgraphs giving component-level traceability of the decision. The framework is evaluated on the MTLP and Abdelhakim datasets with ablation, cross-dataset and edge-deletion robustness tests.

SL. NO	SEARCH RESULT
1. US11470114B2	https://patents.google.com/patent/US11470114B2/en?oq=US11470114B2
2. US9578048B1	https://patents.google.com/patent/US9578048B1/en?oq=US9578048B1
3. US11381597B2	https://patents.google.com/patent/US11381597B2/en?oq=US11381597B2
4. US12413621B2	https://patents.google.com/patent/US12413621B2/en?oq=US12413621B2
5. US20200036751A1	https://patents.google.com/patent/US20200036751A1/en?oq=US20200036751A1
6. US11438377B1	https://patents.google.com/patent/US11438377B1/en?oq=US11438377B1
7. Multimodal Phishing Website Detection System Using XAI (MAKE 2026)	https://doi.org/10.3390/make8010011
8. WebGuard++ (arXiv:2506.19356)	https://arxiv.org/abs/2506.19356

Disclaimer

No legal or statutory compliance advice has been sought. The report and opinion is based on the technical feasibility and practical applications. Only a technical comparison of features with the searched prior art are discussed. The prior art has been performed on third party databases including USPTO, EPO, Delphion, IEEE, ACM and Google patents and the authenticity, accuracy, and validity of information of the results dependent on the accuracy of information listed in these databases. The search does not include results filed before 1976 in USPTO as they are not available in digital form. Furthermore, the performed search does not include results corresponding to patent application filed in USPTO, EPO and others patent office's and that not been published; these patent applicants are not in public domain.